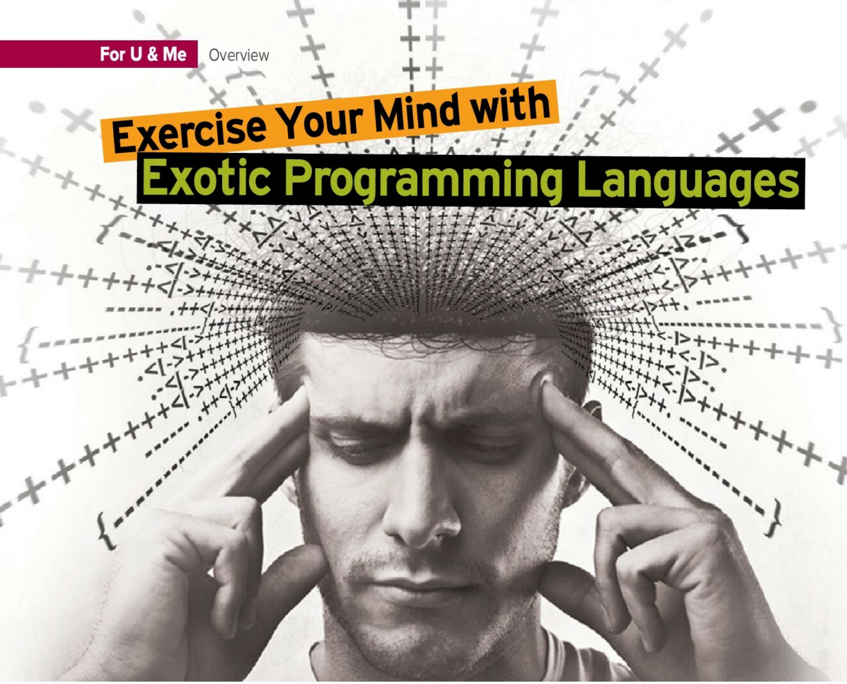
Exercise Your Mind with Exotic Programming Languages



Have you ever encountered Ook! or 'Brainfuck'? No, we are not trying to shock you, but these are actually the names of the exotic programming languages discussed in this article. Learn more about the origin, nature, eccentricities and uses of these languages with a few examples and try-outs.

The word 'exotic' is defined by the Oxford dictionary as "…of a kind not ordinarily encountered." If that is so, what is meant by exotic programming languages? These are languages that are not used for commercial purposes or even for anything 'useful'. This definition can be applied a little loosely and, hence, many different types of programming languages can be classified as 'exotic'. So, after some consideration, I have included three types of languages as belonging to this category. Exotic programming languages include languages that are intentionally designed to be difficult to learn and program with. Such languages are often used to analyse the power of programming languages. They are known as esoteric programming languages. Some exotic languages, known as joke languages, are created for the sake of fun. And the third type includes non-English-based programming languages. The definition clearly mentions that these languages do not serve any practical purpose. Then what's all the fuss about exotic programming languages? To understand their importance, we need to first understand what makes a programming language a programming language.

The 'Turing completeness' of programming languages

A language qualifies as a programming language in the true sense only if it is 'Turing complete'. A language that is Turing complete can be used to represent any computable algorithm. Many of the so called languages are, in fact, not languages, in the true sense. Examples of tools that are not Turing complete and hence not languages in the strict sense include HTML, XML, JSON, etc. But it is often quite surprising to come across Turing complete languages in the most unexpected places. Indeed, it was surprising for me to